From Access to Use: Digital Financial Exclusion among Italian Adults

Evidence from the IACOFI 2023 Survey

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Abstract

This study examines digital financial exclusion in Italy as a two-stage process: first, access to the Internet; second, the adoption and frequent use of online financial services among connected adults. Using 4,862 observations from the IACOFI 2023 survey and the supplied population weights, we combine weighted descriptive evidence with interpretable logistic models. Overall, 89.4% of adults report Internet access, leaving a weighted exclusion rate of 10.6%. Internet exclusion is strongly concentrated among older and less-educated respondents: the unadjusted rate reaches 42.1% among adults aged 70 or more and 62.0% among those with primary education or less. These gradients remain substantial after controlling for gender, work status, macro-region and urbanisation. Among Internet users, 31.1% report having completed at least one financial product or service entirely online, while 75.0% report at least one online financial activity performed often or very often. Adoption follows an inverted-U age profile, peaking among 40–49 year-olds, whereas education displays a monotonic gradient across both adoption and activity. The contrast between QP8 adoption and QP9 activity is substantive: younger adults can be active online without having opened financial products entirely online, while older connected adults use a narrower range of functions. The findings support differentiated policy responses rather than a single binary distinction between “digital users” and “non-users”.

Keywords: Digital exclusion; financial inclusion; Internet access; online financial services; weighted logistic regression; IACOFI 2023.

Dataset: Indagine sulle Competenze Finanziarie (IACOFI) 2023, Banca d’Italia; 4,862 respondents, 219 variables, survey weight `wght`.

AI tool declaration: Generative AI was used to support LaTeX production, editorial structure and language refinement. All empirical values, data transformations and model specifications were reconstructed from the supplied dataset and notebooks.

1 Research Question and Motivation

1.1 Statement

The report addresses one research question:

> **Which demographic groups are less likely to access and use digital financial services?**

The question is divided into two empirical components. The first asks which groups are more likely to lack Internet access. The second asks, among respondents who are connected, which groups are less likely to have completed at least one financial service entirely online and less likely to use online financial functions frequently.

The empirical scope follows the final analytical work contained in the two supplied notebooks: the Internet-access analysis and the QP8–QP9 adoption and activity analysis.

1.2 Motivation

Digital financial inclusion cannot be measured by connectivity alone. Internet access is a necessary gateway, but connected individuals may still avoid online account opening, card applications or digital financial management. Conversely, a person may frequently check an account or pay bills online even if the underlying product was originally opened through a branch. The distinction matters for policy: infrastructure, basic assistance, product onboarding and habitual service use are different barriers and require different interventions.

The main stakeholders are Banca d’Italia, retail banks, FinTech providers, consumer associations and public bodies responsible for digital inclusion. For these actors, the decision problem is not simply how to increase the average adoption rate, but where the process breaks down and which groups require access support, assisted onboarding or simplified service design.

2 Data and Questionnaire

2.1 IACOFI 2023

The analysis uses the IACOFI 2023 microdata, a cross-sectional survey on financial literacy and digital financial skills in Italy. The raw dataset contains 4,862 observations and 219 variables. Population-oriented descriptive estimates use the supplied weight `wght`, in line with the survey documentation and the research proposal [1, 2].

2.2 Questionnaire routing and analytical samples

Three questionnaire blocks are central:

- `qd14`: access to the Internet (Yes/No);
- `qp8_1-qp8_5`: whether the respondent has ever completed five financial activities entirely online;
- `qp9_1`, `qp9_3-qp9_7`, `qp9_10`: frequency of seven online financial activities in the last 12 months.

QP8 and QP9 are administered only when `qd14=1`. Therefore, missing QP8/QP9 values among people without

Internet are structural routing outcomes, not evidence of non-adoption. The analytical design preserves this distinction and never recodes routed missing values as zero.

2.3 Operationalisation

Table 1 summarises the indicators. Special negative codes (for example, "don't know", "not applicable" and refusal) are treated as missing according to the questionnaire. This conservative strategy avoids inflating either exclusion or non-adoption.

3 Empirical Strategy

3.1 Weighted descriptive evidence

For a binary outcome Y_i, the weighted share is

$$\widehat{p}_w = \frac{\sum_i w_i Y_i}{\sum_i w_i}. \tag{1}$$

Uncertainty bands for descriptive rates use a Wilson interval with the Kish effective sample size, $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$ [3]. This is a pragmatic approximation: the available files contain population weights but do not provide the full set of primary sampling unit and stratum identifiers needed for complete design-based variance estimation.

3.2 Stage 1: Internet exclusion model

The first model is a weighted binomial GLM with a logit link:

$$\Pr(NoInternet_i = 1) = \Lambda(\alpha + X_i'\beta), \tag{2}$$

where X_i contains age, gender, education, work status, macro-region and urbanisation. Robust HC1 covariance estimates are used. Reference categories are 18–29 years, man, university education, paid employment, North-West and a city above one million inhabitants. The model includes 4,823 complete observations.

Odds ratios communicate relative associations, while average adjusted predictions translate the same model into probabilities. For each group, its category is assigned to the whole analytical sample while the remaining covariates retain their observed values; predicted probabilities are then averaged using `wght`. Confidence intervals are obtained by simulation from the robust covariance matrix.

3.3 Stage 2: adoption among connected adults

The final QP8 specification is intentionally parsimonious:

$$\Pr(Adoption_i = 1 \mid Internet_i = 1) = \Lambda\big(\gamma + Age_i'\delta + Education_i'\theta\big). \tag{3}$$

It uses 4,249 connected respondents with valid QP8 classifications. The reference groups are age 40–49 and upper-secondary education. Survey weights are normalised to mean one before estimation, and robust HC1 covariance is used. This model is designed to answer the strongest pattern in the QP8/QP9 notebook – the joint role of age and education – without introducing many sparse categorical parameters.

QP9 is analysed descriptively rather than as a second regression outcome. Its role is interpretive: it tests whether low QP8 adoption also means low digital activity. This is essential because QP8 measures product onboarding "entirely online", whereas QP9 measures ongoing operations during the last 12 months.

3.4 Why these visualisations are included

The report deliberately separates visual tasks:

- weighted bar charts show the population-level magnitude of raw gaps;
- adjusted-probability plots show whether age and education differences remain after controls;
- the odds-ratio forest plot summarises the direction and relative size of all significant access correlates;
- paired QP8/QP9 bars demonstrate that adoption and activity are different constructs;
- service-level heatmaps identify which activities drive aggregate differences;
- age-by-education heatmaps reveal compounded disadvantages that marginal summaries can hide.

This sequence mirrors the writing logic of an academic data-science report: description, model-based adjustment, mechanism-oriented decomposition and cautious interpretation.

4 Results

4.1 Overall access, adoption and activity

Weighted Internet access is 89.4%, while 10.6% of Italian adults report no access. Among connected respondents with valid QP8 data, 31.1% have completed at least one financial product or service entirely online. In contrast, 75.0% of connected respondents with valid QP9 data perform at least one online financial activity often or very often.

The gap between 31.1% and 75.0% is not contradictory. QP8 captures a relatively demanding event – opening or subscribing to a product entirely online – whereas QP9 includes routine operations such as checking balances, paying bills and purchasing online. The measures should therefore be interpreted as complementary dimensions of inclusion, not successive rungs of a single deterministic ladder.

4.2 Internet exclusion

4.2.1 Description

Figure 2 shows the weighted unadjusted rates. Internet exclusion is negligible below age 50, then rises sharply: 5.1% at ages 50–59, 19.3% at ages 60–69 and 42.1% at age 70 or more. The education gradient is even steeper: 1.1% among university-educated adults, 4.9% for upper-secondary education, 26.2% for lower-secondary education

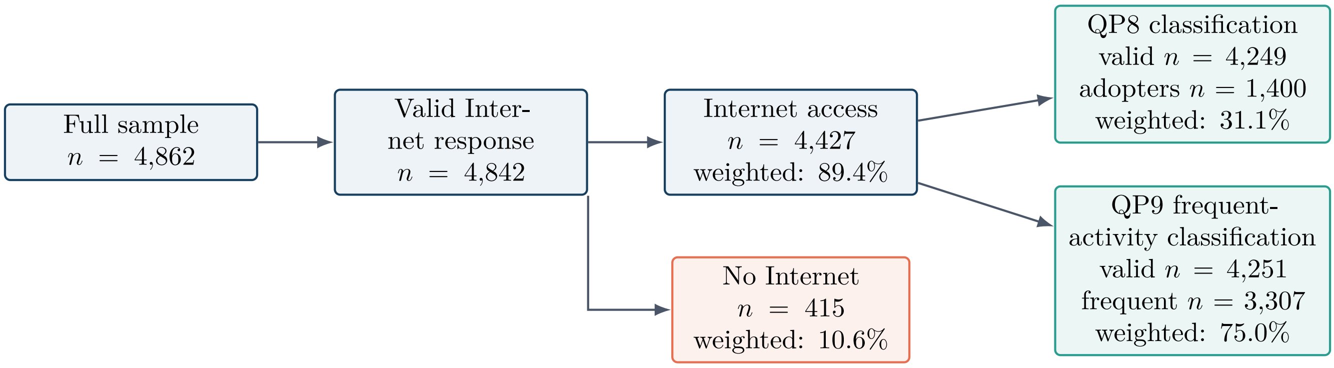

Figure 1: Analytical sample structure. QP8 and QP9 are shown as parallel outcomes among connected respondents, not as a strict funnel. This figure is included to prevent two common errors: classifying people without Internet as non-adopters and assuming that frequent QP9 activity requires prior QP8 online product opening. Counts are unweighted; percentages are weighted.

Table 1: Operationalisation of the core variables

Construct	**Variables**	**Definition and analytical role**
Internet access	`qd14`	1 for a valid Yes, 0 for a valid No, missing otherwise. The access model uses *no Internet* as the outcome so that odds ratios above one directly indicate higher exclusion.
Online financial adoption	`qp8_1`-`qp8_5`	1 if at least one service was completed entirely online; 0 only if all five items are valid and equal to No; missing otherwise. Defined only among Internet users.
Frequent online financial activity	Seven QP9 items	1 if at least one activity was performed Often or Very often; 0 only if every observed item is Never or Sometimes; missing otherwise. Defined only among Internet users.
Demographic controls	`qd1`, `qd2`, `qd3`, `qd7`/`qd7_a`, `qd9`, `qd10`, `qd13`	Gender, macro-region, urbanisation, age, education, work status and income. Age and education are grouped into interpretable categories.
Population weight	`wght`	Used for descriptive rates and weighted GLM estimation.

and 62.0% for primary education or less.

4.2.2 Adjusted probabilities and odds ratios

After multivariate adjustment, the age and education gradients remain substantial (Figure 3). The adjusted probability of no Internet rises from 2.9% for ages 18–29 to 19.3% for ages 70 or more. For education, it rises from 3.0% among university graduates to 30.1% among adults with primary education or less. The difference between raw and adjusted rates indicates that part of the observed disparity is shared with work status, education, region and urbanisation, but the remaining conditional gap is still large.

The forest plot in Figure 4 extends the interpretation beyond age and education. Relative to university education, the odds of no Internet are 7.67 times higher for lower-secondary education and 26.07 times higher for primary education or less. Relative to ages 18–29, the odds are 5.81 times higher at ages 60–69 and 11.46 times higher at age 70 or more. Rural residence is independently associated with higher exclusion (OR 2.83), as are retirement (OR 2.94) and home-care status (OR 2.63). North-East residents show lower odds than the North-West reference group (OR 0.55). Gender is not statistically significant in the multivariate model.

The age 50–59 coefficient is nominally significant (OR 3.06, $p = .025$) but does not survive the notebook's false-discovery-rate adjustment ($p_{FDR} = .062$); it is therefore treated as weaker evidence than the older-age and education effects. A robustness model adding income – including a separate non-disclosure category – leaves the main gradients materially unchanged: the ORs remain 12.73 for age 70+, 20.34 for primary education or less and 3.06 for rural residence.

Table 2: Selected results from the weighted Internet-exclusion model

Factor	**OR**	**95% CI**	p
Age 60–69	5.81	2.17–15.58	$< .001$
Age 70+	11.46	4.18–31.41	$< .001$
Upper secondary	2.26	1.23–4.15	.009
Lower secondary	7.67	4.24–13.88	$< .001$
Primary or lower	26.07	12.93–52.56	$< .001$
Retired	2.94	1.90–4.54	$< .001$
Rural area	2.83	1.45–5.52	.002
North-East	0.55	0.39–0.78	.001

4.2.3 Interpretation

Internet exclusion is not evenly distributed across the population. It is primarily an age-and-education problem, reinforced by rural location and life-course status. The adjusted results imply that older age is not merely a proxy for lower education or retirement: an independent gap

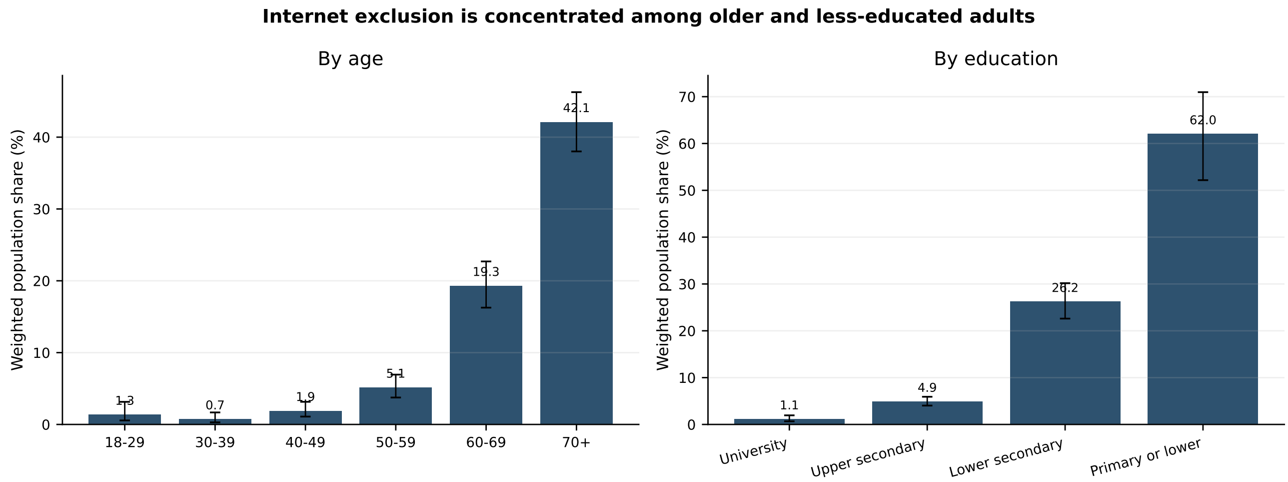

Figure 2: Weighted Internet-exclusion rates with approximate 95% confidence intervals. The figure is included to display the absolute scale of the digital divide before statistical adjustment.

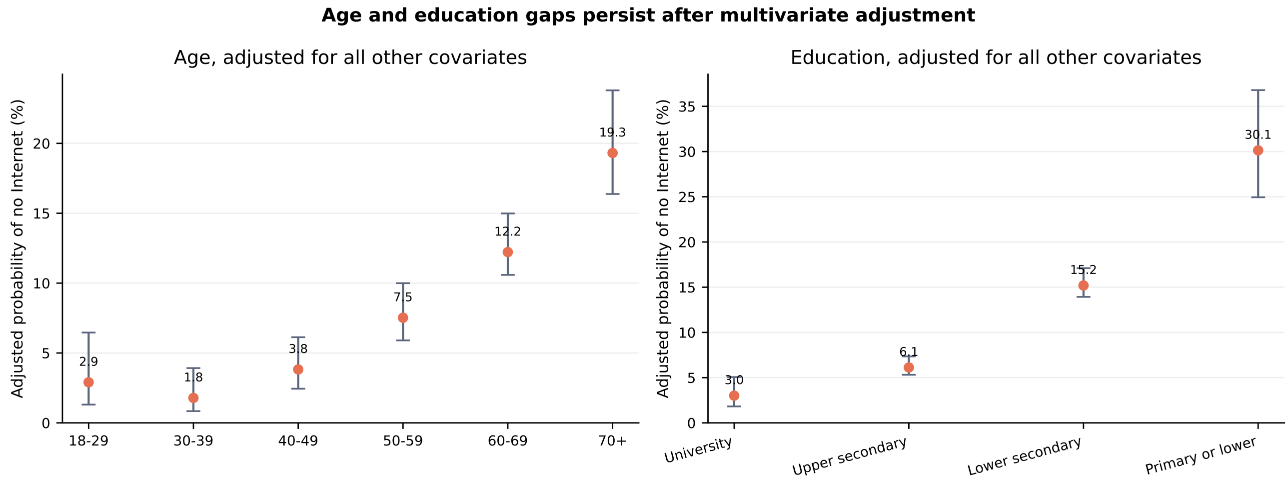

Figure 3: Average adjusted probabilities of no Internet, derived from the weighted multivariate GLM. This figure is included because probabilities are more interpretable for policy than log-odds and allow direct comparison with the descriptive rates.

remains after these characteristics are held constant. At the same time, the reduction from a raw 42.1% exclusion rate among people aged 70+ to an adjusted 19.3% probability shows that policy should not attribute all of the age gap to age itself. Education, employment history and geography form part of the same vulnerability structure.

4.3 Online adoption and frequent activity among Internet users

4.3.1 Aggregate demographic patterns

Figure 5 compares two outcomes. QP8 adoption follows an inverted-U age pattern: 20.5% at ages 18–29, 32.0% at 30–39, 41.1% at 40–49, 36.6% at 50–59, 26.6% at 60–69 and 19.3% at 70+. Frequent QP9 activity is much higher in every group, peaking at 87.7% among 30–39 year-olds and declining to 49.1% at age 70+.

Education produces a monotonic gradient across both measures. Adoption rises from 15.7% among respondents with lower-secondary education or less to 31.5% with upper-secondary education and 41.4% with university education. Frequent activity rises from 49.0% to 77.2% and 88.8%, respectively.

The young-adult result is particularly important. Ages 18–29 have low QP8 adoption (20.5%) but substantially higher frequent QP9 activity (69.9%). It would therefore be misleading to label all young non-adopters as digitally excluded. The pattern may reflect life-cycle demand, lower product ownership or fewer occasions to open insurance, credit and investment products, rather than a lack of operational digital familiarity.

4.3.2 Which services drive adoption?

The QP8 heatmaps in Figure 6 show that aggregate adoption is mainly driven by account opening, card requests and insurance. Among 40–49 year-olds, the weighted rates are 21.0% for online account opening, 22.8% for card requests and 18.6% for insurance. Credit and platform investment remain much less common in every age group. The education gradient is broad rather than product-specific: university graduates report 21.3% for account opening, 25.2% for card requests and 21.6% for insurance, compared with 8.9%, 7.9% and 5.3% among respondents with lower-secondary education or less.

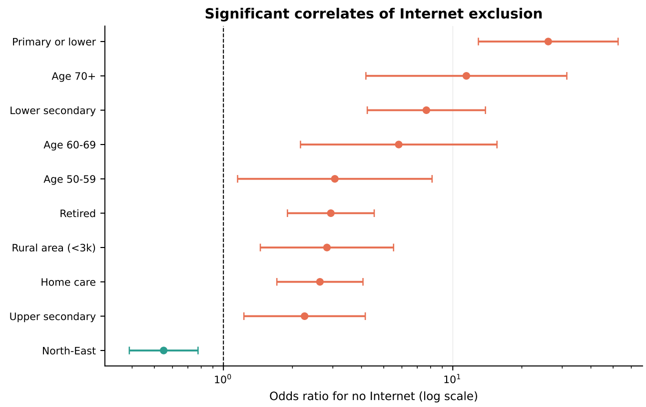

Figure 4: Significant odds ratios from the Internet-exclusion model. Values above one indicate higher odds of no Internet; values below one indicate lower odds. The log scale permits effects of very different magnitudes to be shown in one panel.

4.3.3 Which online activities are frequent?

QP9 reveals a different hierarchy (Figure 7). Checking balances and transactions is the most frequent activity, reaching 78.4% among ages 30–39 and 79.9% among university graduates. Paying bills and purchasing online are also widespread. Mobile payments, managing financial products and robo-advice are less frequent, but still show clear education gradients.

Connected older adults are not uniformly inactive. Among respondents aged 70+, 34.2% frequently check balances and transactions and 31.5% frequently pay bills online. The policy implication is that older adults who have crossed the access barrier may retain meaningful digital capability, although their range and intensity of activities are narrower.

4.3.4 Joint age and education patterns

Figure 8 combines age and education. The strongest QP8 adoption is observed among university-educated adults aged 40–49 (51.1%) and 50–59 (48.9%). QP9 frequent activity exceeds 90% for university-educated adults aged 30–59. At the opposite end, adults aged 70+ with lower-secondary education or less show 9.0% QP8 adoption and 30.8% frequent QP9 activity.

These interaction cells are descriptive, not causal. Several young, low-education cells are small (for example, 38 connected respondents aged 18–29 with lower-secondary education or less), so individual cell values should not be overinterpreted. The heatmap is included to reveal compounded patterns while making this sparsity visible through the accompanying sample counts in the reproducibility tables.

4.3.5 Adjusted adoption probabilities

Mutually adjusting age and education preserves the inverted-U age profile (Figure 9). The adjusted probability of QP8 adoption is 19.9% at ages 18–29, 29.1% at 30–39, 39.2% at 40–49, 36.8% at 50–59, 29.9% at 60–69 and 24.0% at 70+. Compared with ages 40–49, the odds are significantly lower for 18–29 (OR 0.37), 30–39 (OR 0.63), 60–69 (OR 0.65) and 70+ (OR 0.48); ages 50–59 are not significantly different (OR 0.90, $p = .307$).

The education gradient remains strong after age adjustment. The predicted adoption probability is 16.3% for lower-secondary education or less, 31.4% for upper-secondary education and 40.8% for university education. Relative to upper-secondary education, the odds are 58% lower for lower-secondary education or less (OR 0.42) and 52% higher for university education (OR 1.52).

Table 3: Weighted adoption model among Internet users

Factor	**OR**	**95% CI**	p
Age 18–29	0.37	0.29–0.48	$< .001$
Age 30–39	0.63	0.51–0.77	$< .001$
Age 50–59	0.90	0.74–1.10	.307
Age 60–69	0.65	0.52–0.82	$< .001$
Age 70+	0.48	0.36–0.65	$< .001$
Lower secondary or less	0.42	0.34–0.52	$< .001$
University	1.52	1.30–1.77	$< .001$

References: age 40–49; upper-secondary education.

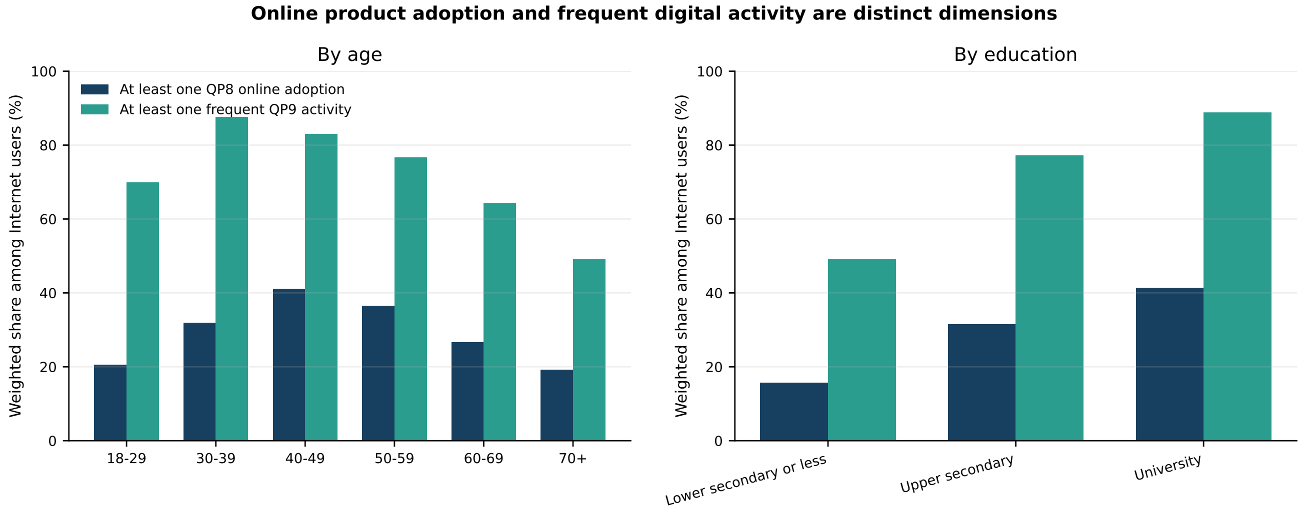

Figure 5: Weighted QP8 adoption and QP9 frequent activity among Internet users. The side-by-side design is included to demonstrate that low online product onboarding does not necessarily imply low routine digital activity.

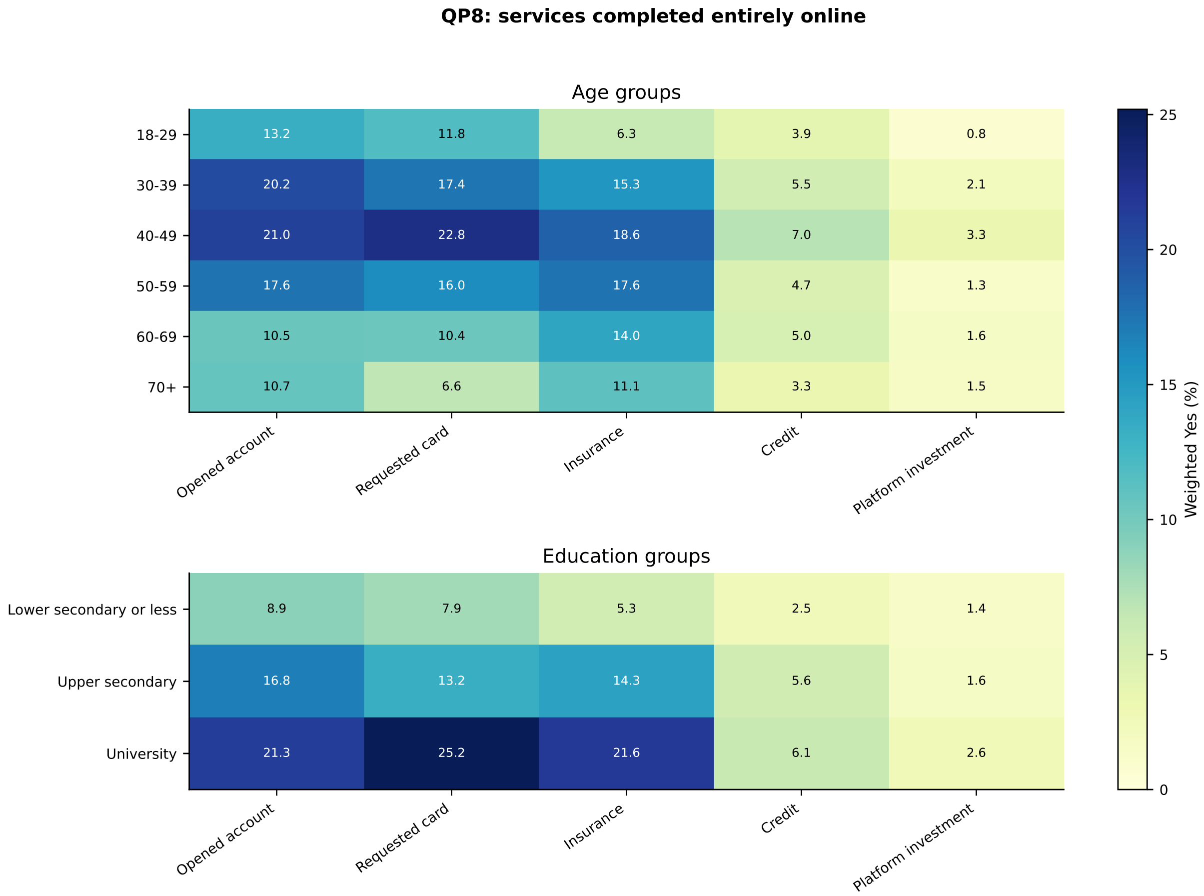

Figure 6: Weighted percentages answering Yes to each QP8 item. This decomposition is included because a single adoption indicator can conceal whether gaps concern basic transactional products or more specialised products.

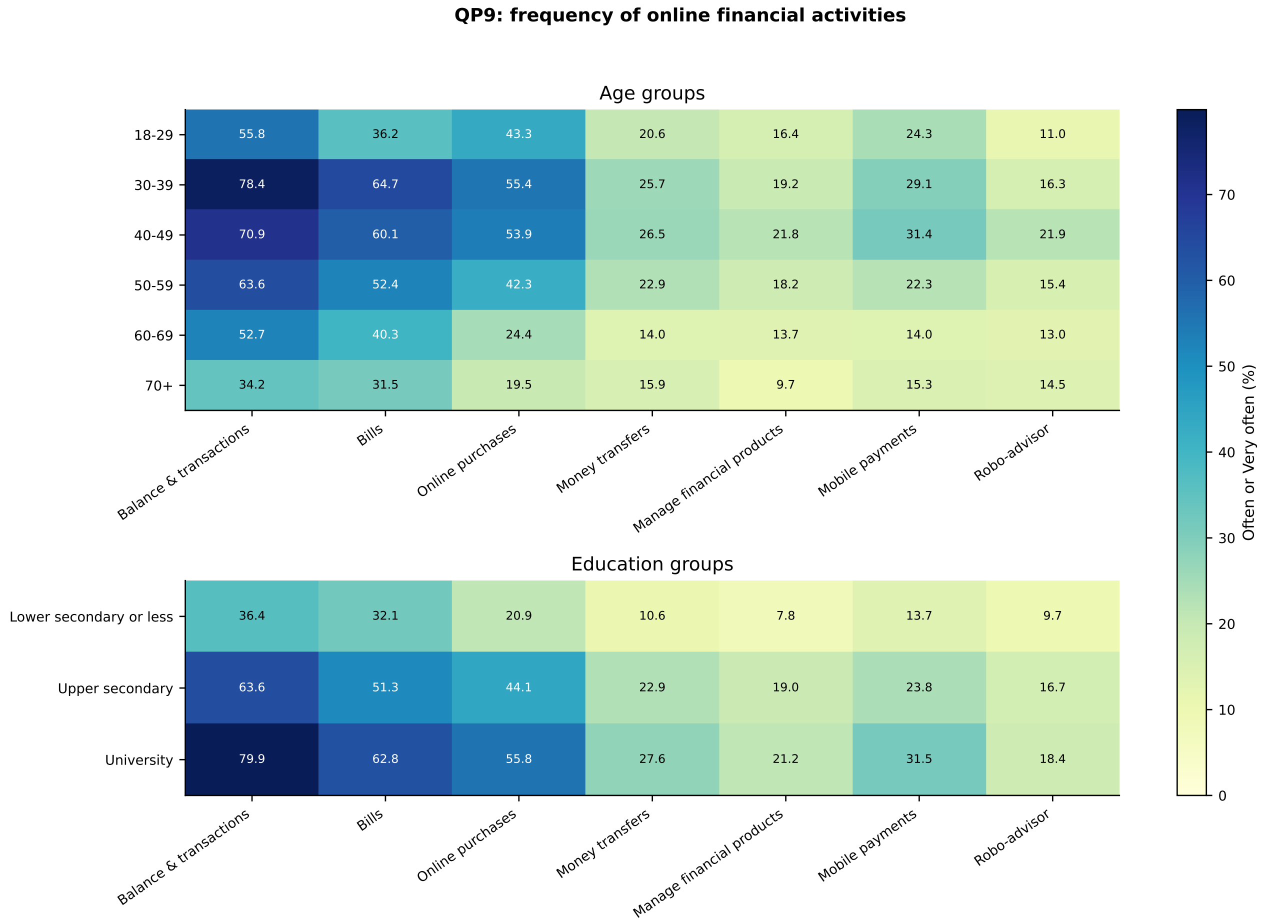

Figure 7: Weighted percentages reporting Often or Very often for each QP9 activity. The heatmap is included to identify the operational functions behind the composite frequent-activity indicator.

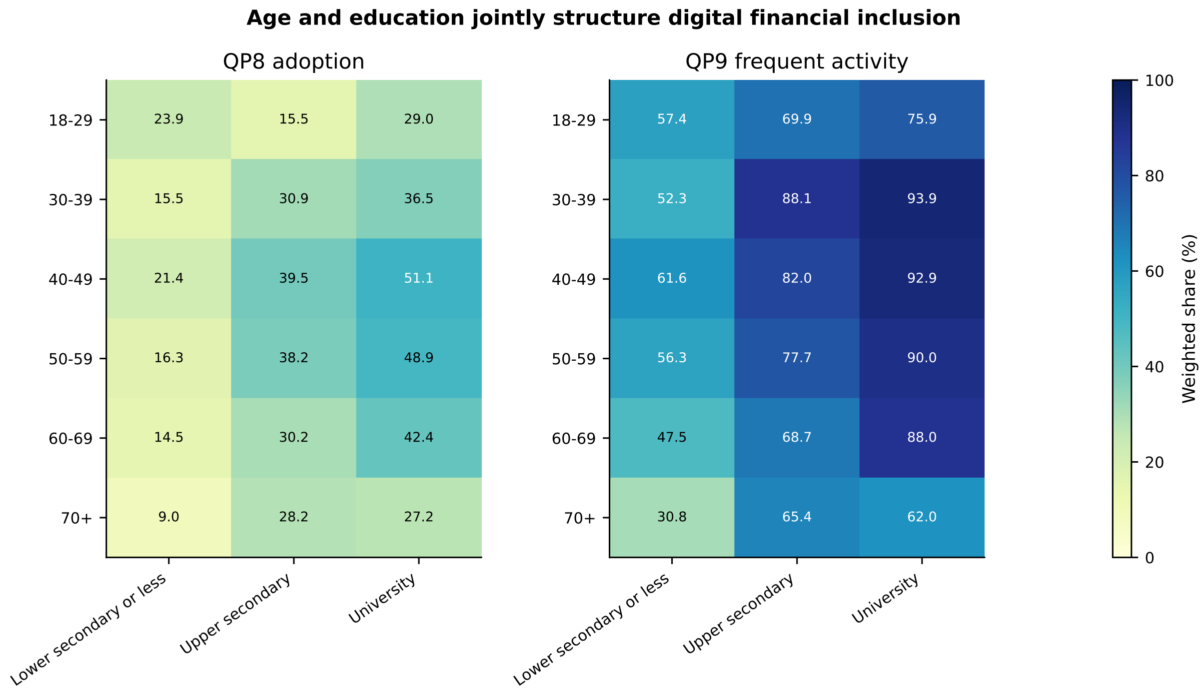

Figure 8: Weighted QP8 adoption and QP9 frequent activity by age and education. The figure shows how the two stratifiers combine rather than acting only as separate marginal effects.

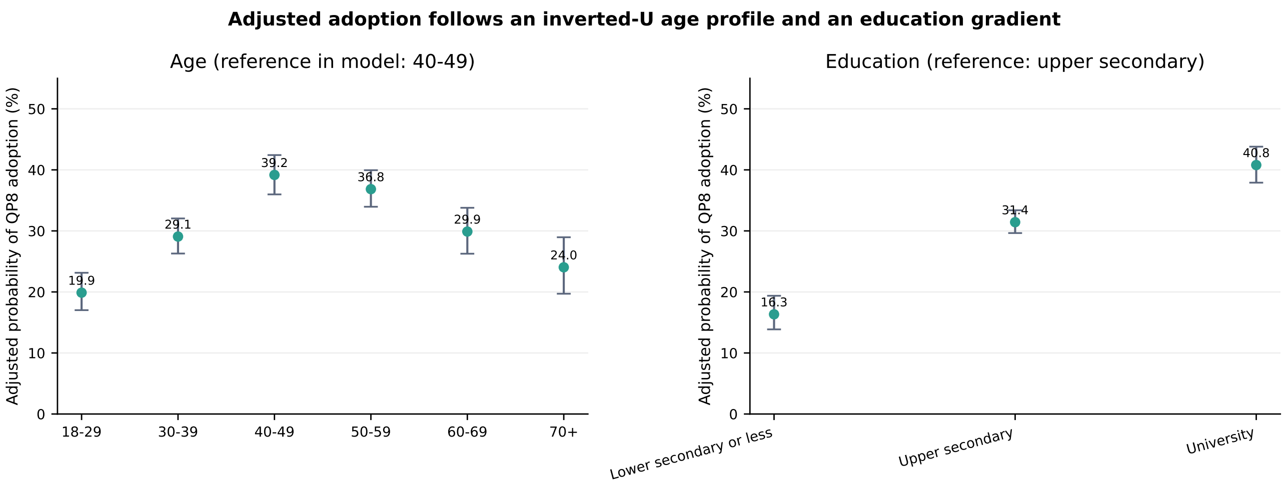

Figure 9: Average adjusted probabilities of QP8 adoption from the weighted age-and-education GLM. This figure is included to distinguish the non-linear life-cycle pattern from the monotonic education gradient.

5 Discussion and Policy Implications

5.1 A two-layer digital divide

The findings reveal two different forms of exclusion. The first is a hard access divide, concentrated among older adults, people with low education and rural residents. The second is a use-and-onboarding divide among people who are already connected. Education remains strongly associated with both QP8 adoption and QP9 activity, suggesting that connectivity alone does not equal practical financial inclusion.

Age operates differently at the two stages. For Internet access, the relationship is strongly monotonic: exclusion rises sharply in older groups. For online product adoption, the relationship is inverted-U, with the highest probability in middle age. This difference is substantively coherent because product acquisition is shaped by financial life-cycle needs. It also demonstrates why a single model that classifies everyone without QP8 adoption as "digitally excluded" would be analytically misleading.

5.2 Implications for stakeholders

Three intervention profiles emerge:

- **Structurally unconnected adults.** Older, low-education and rural groups require access support, assisted digital services and the continued availability of non-digital channels.
- **Connected but weakly engaged adults.** Simplified interfaces, guided onboarding, transparent product information and branch-to-digital transition support are more relevant than infrastructure alone.
- **Connected active users with limited QP8 adoption.** Particularly among younger adults, lower product opening may reflect demand and life-cycle conditions. Interventions should avoid treating non-adoption as incompetence and instead focus on relevance, affordability and financial needs.

A practical policy dashboard should therefore report at least three indicators separately: Internet access, online product onboarding and frequent digital financial activity. Aggregating them into a single score would hide the distinct age profiles and could lead to poorly targeted interventions.

6 Limitations

- **Cross-sectional design.** The models identify associations and do not establish causal effects.
- **Self-reported outcomes.** Access, adoption and activity can be affected by recall or interpretation errors.
- **Different time frames and constructs.** QP8 asks whether an activity has ever been completed entirely online, while QP9 refers to frequency during the last 12 months. They are not interchangeable measures.
- **Demand and life-cycle confounding.** A lower QP8 adoption rate may reflect less need for credit, insurance or investment products rather than digital incapacity.
- **Survey-design uncertainty.** Weighted point estimates are available, but the absence of PSU and stratum variables prevents full design-based standard errors. The descriptive confidence intervals are therefore approximations.
- **Sparse interaction cells.** Some age-by-education combinations have small sample sizes and should be read as exploratory.
- **Income non-response.** Income is excluded from the main access model to avoid losing a large portion of the sample. A specification with an explicit non-disclosure category produces similar main results, but non-random missingness remains possible.
- **Parsimonious adoption model.** The final QP8 model focuses on age and education; unmeasured factors such as product ownership, perceived usefulness, trust and prior branch relationships may explain additional variation.

7 Conclusion

Digital financial exclusion in Italy is best understood as a sequence of distinct barriers rather than a binary user/non-user classification. The first barrier is Internet access, where older age, low education and rural residence are the dominant correlates. The second barrier concerns the form and intensity of online financial participation among connected adults. Education remains a consistent divider, while age becomes non-linear: middle-aged adults are most likely to have completed financial products entirely online, and younger adults often display frequent activity without corresponding product onboarding.

The central empirical lesson is that access, adoption and activity answer different questions. Policy that focuses only on Internet connectivity misses substantial inequalities among connected adults; policy that focuses only on online product adoption risks misclassifying active young users as excluded. A differentiated framework – preserving access support, simplifying onboarding and measuring actual activity separately – provides a more accurate basis for digital financial inclusion strategies.

Reproducibility Materials

The project folder accompanying this report contains the LaTeX source, the analysis-reproduction script, all figures in PDF and PNG format, and CSV tables containing the exact values used in the text. The source is structured so that author names, title elements and figure selection can be edited without changing the analytical pipeline.

References

[1] Banca d'Italia. *Indagine sulle competenze finanziarie degli adulti in Italia (IACOFI) 2023*. Rome, 2023.

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